

ABSTRACT ALGEBRA 1 - HOMEWORK 8- M. VICARÍA

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you **SHOULD JUSTIFY COMPLETELY YOUR ANSWER** with a proof or a counter-example. No credit is granted to solutions without justification.

1. MANDATORY PROBLEMS

Problem 1:(1 point) Let p be a prime number and H be a group. Prove the following:
 If $\phi : \mathbb{Z}/p\mathbb{Z} \mapsto H$ is a homomorphism, then either ϕ is trivial (i.e. $\text{Img}(\phi) = \{e_G\}$) or injective.

Problem 2: (1 point) Let N be a normal subgroup of G . If $(G : N) = 7$ determine all subgroups $H \subseteq G$ such that $N \subseteq H$.

Problem 4:**(4 points) Let $B_3(\mathbb{R})$ be the group of 3×3 matrices with multiplication that are upper triangular and invertible.

- (1) Let $U = \left\{ \begin{bmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\}$. Show that U is a normal subgroup of $B_3(\mathbb{R})$.
- (2) Let $G = \mathbb{R}^3$ with the following operation $\star$: $[a_1, a_2, a_3] \star [b_1, b_2, b_3] = [a_1 \cdot b_1, a_2 \cdot b_2, a_3 \cdot b_3]$. Then $(G, \star)$ is a group. Prove that $B_3(\mathbb{R})/U \cong G$.
- (3) Let $H_1 = \left\{ \begin{bmatrix} 1 & 0 & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \mid b, c \in \mathbb{R} \right\}$. Show that H is a normal subgroup of U and prove that $U/H_1 \cong (\mathbb{R}, +)$.
- (4) Let $H_2 = \left\{ \begin{bmatrix} 1 & 0 & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mid c \in \mathbb{R} \right\}$. Show that H_2 is normal in H_1 , and prove that $H_2 \cong (\mathbb{R}, +)$ and $H_1/H_2 \cong (\mathbb{R}, +)$.

Problem 5:**(4 points) A group is said to be simple if the only normal subgroups of G are G and $\{e_G\}$.

- (1) Prove that $\text{Alt}_n \mapsto \text{Sym}_n$ is a normal subgroup. How many distinct conjugacy classes of Sym_6 are contained in Alt_6 ?
- (2) Prove that $\text{Sym}_n/\text{Alt}_n \cong \text{Sym}_2$.
- (3) Show that there are no normal subgroups between Alt_n and Sym_n .
- (4) Prove that the alternating group Alt_n is simple.